TEJAS N

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PROFESSIONAL SUMMARY

Third-year B.Tech Computer Science student at MIT Bengaluru specializing in AI/ML engineering and full-stack web development. Hands-on experience building production-ready AI applications including RAG systems, LLM-powered tools, and data analytics platforms using Python, React.js, LangChain-Graph, and Hugging Face Transformers.

EDUCATION

B.Tech, Computer Science and Engineering **August 2024 - May 2028**

Manipal Institute of Technology, Bengaluru | CGPA: 7.33 / 10.0 (up to 4th Semester)

Relevant Coursework: DSA, Database Management Systems, Machine Learning, OOPS, Operating Systems, Computer Networks

TECHNICAL SKILLS

Languages:	Python, JavaScript, C, Java, HTML5, CSS3
Frontend:	React.js, Next.js, Tailwind CSS, Streamlit
Backend & DB:	FastAPI, Node.js, REST APIs, PostgreSQL, PostGIS, SQLite, Firebase
AI / ML & Agents:	PyTorch, Scikit-learn, FAISS, Hugging Face Transformers, LangChain, LangGraph, OpenCV
Developer Tools:	Git, GitHub, Docker, Vercel, n8n, CI/CD, VS Code
Concepts:	RAG, AI Agents, NLP, Semantic Search, Transformer Architecture, Data Pipelines, Full-Stack Dev

PROJECTS

SATQ AI - Geospatial Foundation Platform Sep 2026 onwards - SIH Hackathon

React.js, FastAPI, PyTorch, Gemini 2.5 Flash, Grounding DINO, SAM 2, PostGIS, OpenCV, Leaflet.js

- Built an agentic platform using Gemini 2.5 Flash as an AI orchestrator to parse natural language queries, cutting spatial analysis time from 4+ hours to
- Engineered dynamic ROI engine with Leaflet.js and OpenCV streaming 512x512 tile crops, reducing GPU compute overhead by >70%.
- Integrated multimodal AI pipeline (Grounding DINO, SAM 2, Prithvi-EO, Siam-Swin ViT) for zero-shot segmentation, change detection, and auditable C

CDS - AI-Powered Crack Detection System Hackverse 2.0 - IBM | Aug 2026

React.js, TypeScript, FastAPI, OpenCV, OpenRouter, Google Gemini, IBM Granite, SQLite

- Built an intelligent infrastructure inspection platform supporting image, video, and live-camera crack detection with automated severity assessment and
- Designed a multi-provider AI pipeline using OpenRouter, Gemini, and OpenCV fallback for resilient defect detection and structured bounding boxes.
- Selected for the final round among 90 teams at Hackverse 2.0, an IBM-judged hackathon.

AI-Powered RAG Question Answering System Jan 2025 - Mar 2025

Python, FAISS, Hugging Face Flan-T5, LangChain, Streamlit, NLP, Semantic Search

- Developed a RAG pipeline enabling context-aware Q&A over custom document corpora using FAISS vector retrieval, Flan-T5, and LangChain; sub-50
- Implemented document chunking and semantic embedding strategies, reducing document lookup time by ~70% across 100+ page corpora.
- Designed an interactive Streamlit interface supporting dynamic document ingestion and real-time grounded query processing.

SOUNDMATCH - Parallel Audio Fingerprinting & Song Identification Systems & High Performance Computing

C, OpenMPI, Python, Librosa, NumPy, SciPy, Flask, HTML5, CSS3, JavaScript

- Extracted time-frequency landmarks from audio signals, generating robust combinatorial fingerprints for song matching.
- Distributed database search across multiple processes using OpenMPI, achieving significant speedup over sequential search benchmarks.

CERTIFICATIONS

* IBM - Fundamentals of AI Agents Using RAG and LangChain
* Meta - Front-End Development
* Microsoft - Data Analysis and Visualization in Python
* BCG X - GenAI Job Simulation | AI-powered financial chatbot development